



Exploring adaptations of groundnut cropping to prevailing climate variability and extremes in Limpopo Province, South Africa

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ABSTRACT

Adapting groundnut production to the high climatic variability and extremes prevailing in Limpopo (South Africa) requires season- and site-specific exploitation of genotype x environment x management interactions. Notably, the El Niño-Southern Oscillation phenomenon (ENSO) causes distinctive seasonal rainfall anomalies in the province: Weather records over the period 1955–2010, showed El Niño years with an average of 296 mm growing season rainfall, neutral years with 349 mm and La Niña with 434 mm. We hypothesized that (i) early planting date is crucial for ensuring high productivity of groundnut and (ii) early maturing cultivars can ensure satisfactory production even in El Niño years, both options aiming at escaping prolonged periods of drought. To evaluate our hypotheses, we conducted a three year on-station trial with three planting dates and two cultivars, a one year trial at two sites comparing the two cultivars, tested crop model APSIM against field trial data, and, finally, performed long-term gridded crop model simulation runs for all cultivated land (i.e. 14,739 grid cells at a resolution 0.01°) in Limpopo exploring the effect of the cultivars and planting dates on groundnut yield. Field trials showed that early planting was superior to later plantings, irrespective of the weather year. Even in the very dry El Niño year 2016, yields of more than 1000 kg/ha were achieved with early planting, while planting later was not possible due to severe soil water deficits. Cultivar ICGV 03796 performed slightly better than ICGV 99257 due to faster ground cover closure, but no difference in phenology was observed. After evaluation of the model against the field trial data was found satisfactory, APSIM simulations showed that highest yields were achieved in La Niña years (median 996 kg/ha across all grids) and smallest in El Niño years (654 kg/ha across all grids). However, as in the field trials, we observed on average over all grid cells higher grain yields for early planting for all weather years as effective rainfall captured was higher than for later plantings. Longer growth duration by cultivar ICGV 03796 appeared to be more productive than an earlier maturing cultivar that we added into the simulation analysis as a third cultivar. However, a closer look at the spatial pattern of the simulation analysis revealed that for the generally drier regions of Limpopo, this cultivar with shorter growth duration would indeed be more productive in El Niño years. It showed that such season-specific adaptation would increase yields only by 2.9% for Limpopo province in total, but yield gains of more than 10% could be obtained for 11.3% of the grid cells in El Niño years. Hence, this study demonstrated that management recommendations at provincial scale explicitly need to consider in more detail a site- and season-specific approach.

1. Introduction

Groundnut (*Arachis hypogaea* L.) has been an important cash crop for smallholder as well as for commercial farming systems in South Africa in the 1960ies and 70ies with average annual production of 300,000 tons (FAOSTAT, 2017). However, since then groundnut

production declined to roughly 50,000 tons, in the 2010s. The lowest figure was reached in 2016 with an estimated production of 20,000 tons. This turned South Africa into an importer of 30,000–50,000 tons to fulfil the country's demand. While the high demand for labor and lack of improved genetic material are perceived as long-term reasons for the decline in production, in many parts of South Africa, and

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especially in Limpopo province, adaptation to climate variability is a major challenge for successful groundnut farming (Conway et al., 2015). In particular, the already high rainfall variability might even get exacerbated under climate change (Conway et al., 2015). The low production in 2016 was driven by the El Niño-Southern Oscillation (ENSO) phenomenon, which in certain years has caused severe dry spells in the region. Such climate extreme events, triggered by ENSO, can be predicted for the next season with a fairly high confidence for this region (Malherbe et al., 2014).

However, groundnut management needs to be well adapted to such variable conditions; i.e. it requires an efficient exploitation of season- and site-specific environment $\times$ management $\times$ genotype ($G \times M \times E$) interactions to revitalize growth in production. Keeping groundnut in crop rotations would have additional benefits apart from the production aspect, which are nitrogen fixation (and increased soil fertility), and diversification of the narrow range of crops found in the predominating maize-based systems of the region. Hence, these are some good reasons why agronomists currently focus on identification of promising legume genotypes and their management, fitting well to local soil and climate conditions (Farrow et al., 2016).

While carrying out field trials is a classic tool for agronomists to understand crop performance for a given environment, they are expensive and resource demanding. Thus, it is hardly possible to run such trials for several years and sites, which are required to develop season and site specific recommendations for a wide range of $G \times M \times E$. Process-based crop modelling has evolved as an additional option to identify production limitations and adapted cropping practices. For instance, Hoffmann et al. (2017) used the model APSIM (Agricultural Production System sIMulator) to demonstrate the benefits of zone and season specific fertilizer management in low rainfall south-east Australia by analyzing soil-weather-plant interactions. Models like APSIM simulate on a daily basis crop growth in dependence of available resources light, water, temperature and nitrogen (Holzworth et al., 2014). While this makes it data demanding (climate & soil) in terms of running, it enables it at the same time to investigate $G \times M \times E$ interactions at a relatively detailed level, including in-season rainfall distribution, which is crucial for understanding growth patterns in the highly variable climate of Limpopo. Key requirement for crop model application is prior model evaluation for the region. This has been done for many crop models intensively for Europe, America, Asia and Australia (for instance: Holzworth et al., 2014; Jones et al., 2003), but there are few tested models for southern Africa. APSIM is an exception, having its roots in East Africa, but also used intensively in southern Africa (for an overview, see Whitbread et al., 2010). However, the usage of crop models cannot replace field trials. On the contrary, in recent years there is growing awareness, especially in the crop modelling community, that good field trials have become scarce, but are crucial to improve and test the models (Rötter et al., 2011). In addition, crop models are not well prepared yet to deal with all yield-constraining factors, which farmers face, notably the lack of capability to simulate pest and diseases. Having this in mind, we combined on-station field trials with crop modelling to test two management options across Limpopo, the most vulnerable province of South Africa in terms of climate variability and food security (Conway et al., 2015; De Cock et al., 2013). We hypothesized that (i) early planting date is crucial to ensure high productivity of groundnut and (ii) short-term cultivars ensure satisfying production even in El Niño years. Both options are based on the idea to escape prolonged periods of drought by earlier harvesting – and among the frequently proposed options as an adaptation strategy to terminal drought in Africa. However, MacCarthy et al. (2017) showed for northern Ghana a much more complex response to shifting planting date, indicating that this might not be the case in every region. Nevertheless, for southern Africa Shumba et al. (1992) showed that for maize one day delay in planting of maize in Harare region (Zimbabwe) during the period of 5-October to mid of December resulted in 1.35% lower yields. This is in line for other regions in southern Africa

(MacColl, 1990). Similarly, sunflower yields have been shown to be reduced in South Africa due to late planting (Ma'ali, 2017). Consequently, also the recommendation for groundnut in South Africa is to plant early. Besides planting date, also cultivar choice is seen as an important adaptation to the local climate conditions. Makaudze (2014) found early maturing cultivars in maize beneficial in El Niño years in Zimbabwe. The information from these crops served as basis for developing our hypotheses.

To evaluate our hypotheses we went through the following steps: (i) we conducted a three year on-station trial with three planting dates and two cultivars, (ii) carried out a one year trial at two sites comparing the two cultivars, (iii) evaluated APSIM against the trial data, and finally (iv) performed long-term grid-based crop model simulation runs for all arable land in Limpopo investigating the performance of cultivars and the effect of planting date across the province. This combination of methods should build the basis for arriving at improved management recommendations to better adapt groundnut production to climate variability and extremes.

2. Material & methods

The study took place in the Limpopo province, South Africa. It borders Mozambique, Zimbabwe and Botswana. Overall climate conditions differ spatially in the province and ranges from warm-desert climate (BWh according to Koeppen classification) in the West, to warm-semi-arid (BSh) to humid subtropical climate (Cwa) in the East of the province (Engelbrecht and Engelbrecht, 2016). Rainy season occurs during summer (October–April) (see Supplement Fig. A–C). The winter period is characterized by extended dry spells. The average growing rainfall differs substantially in the province, from 320 mm growing season to more than 600 mm over the period 1955–2010 (Kalnay et al., 1996; Sheffield et al., 2006). In addition to the strong spatial variability, a strong difference between seasons exists in the region (Fig. 1), which is caused a.o. by ENSO (Moeletsi et al., 2011; Reason et al., 2005). The categorization of the years into El Niño, La Niña and neutral is usually based on the Oceanic Niño Index (ONI) (NOAA, 2017). Running 3-month mean ONI values for the months December, January, and February representing the core months of the cropping season were used in our case to capture the core period of the growing season. If value was above one, it was classified as El Niño, if below minus one as La Niña, and in-between as neutral years. This resulted for the period from 1955 to 2010 in eight El Niño years and seven La Niña years (see Supplement Fig. D). Indeed, rainfall during the growing season (October-to April) was different as expected; El Niño (296 mm), neutral years (349 mm), and La Niña (434 mm) when aggregated over the whole province.

Soil water holding capacity varies widely across the province from 20 to 140 mm, but remains overall low (Fig. 2). Limpopo is economically the least developed province of South Africa and has also a high share of smallholder farms in comparison to commercial farms (Schoeman et al., 2013).

2.1. Field trials

2.1.1. Time of planting $\times$ crop cultivar trials

The trials were carried out in the cropping seasons 2013/14, 2014/15 and 2015/16 at the experimental farm of the University of Venda (Univen) in Thohoyandou (Fig. 1). The soil is clay with a low soil organic carbon content of 0.4–0.7% in the top soil. The experiment was laid out using a randomised complete block design in factorial arrangement with three replicates: treatments were planting date (early, mid and late) and cultivar. In the first year, planting dates were 09/12/2013 (early), 15/01/2014 (mid) and 09/02/2014 (late), in the second year 02/12/2014 and 03/01/2015 and in the third year 5/12/2015 and 30/01/2016. For third planting date, sowing was not possible in the seasons 2014/15 and 2015/16 due to severe drought. Two cultivars

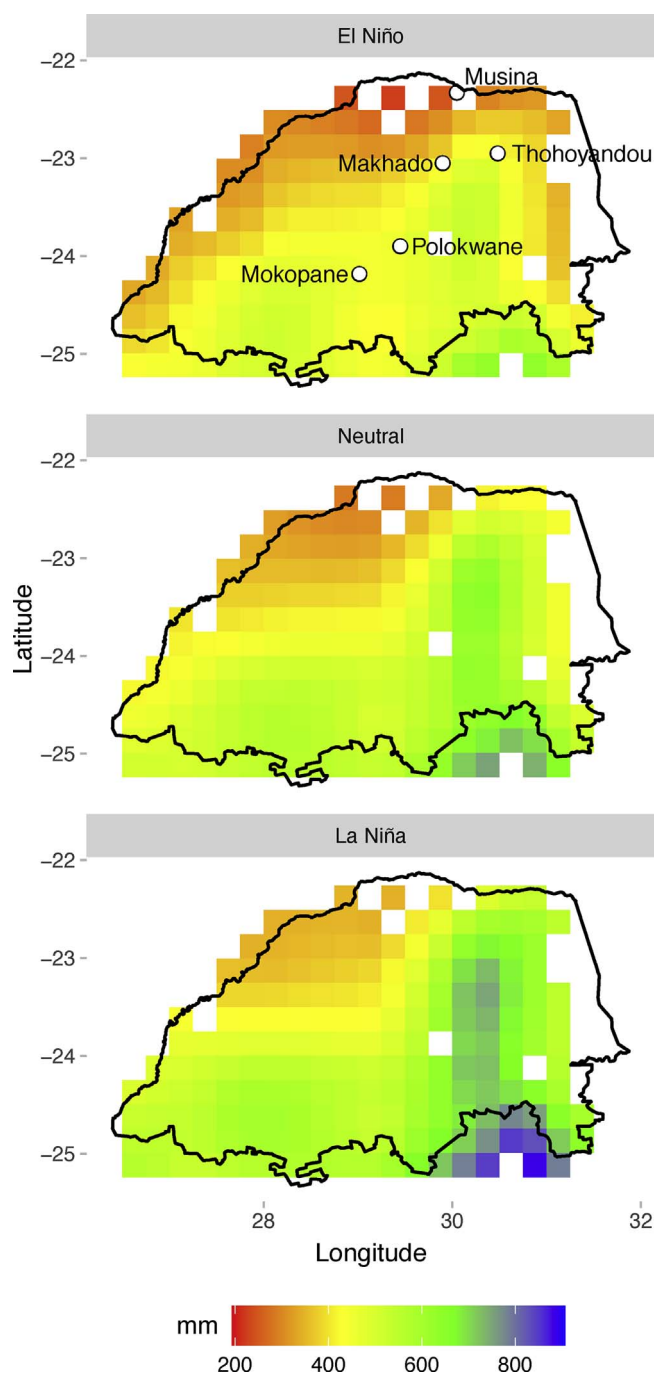


Fig. 1. Average rainfall during the growing season (October–April) for Limpopo for El Niño, neutral and La Niña years according to Sheffield et al. (2006) and Kalnay et al. (1996). See Section 2.3.2 for more details.

were selected from the ICRISAT groundnut breeding program: ICGV 07396 and ICGV 99257. Both cultivars belong to the so called Spanish group of groundnut, which have typically high oil content. This group is dominant in South Africa (Department of Agriculture, 2010). Cultivar ICGV 07396 was considered as drought tolerant and the ICGV 99257 as a short-duration cultivar. Maturity took place at the experimental research station of ICRISAT in Patancheru (India) for ICGV 07396, 110–120 days after sowing, and for the ICGV 99257, after 95–100 days.

The plot size was 2 m × 2.5 m. Planting density was 6.67 plants/m², which was lower than the standard recommendation of 13–15 plants/m² for South Africa (Department of Agriculture, 2010). However, this recommendation is rather vague taking into account the wide range of

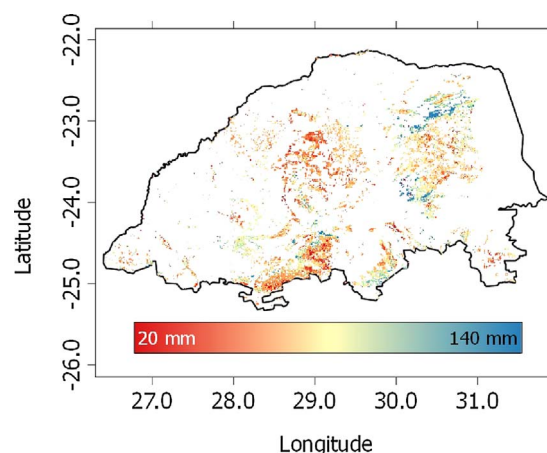


Fig. 2. Total plant available soil water holding capacity limited to cropping land according to South Africa National Land-Cover Change Map 2000 (Schoeman et al., 2013) and soil dataset RZ-PAWHC (Leenaars et al., 2015).

climate conditions found in South Africa. Limpopo is on average one of the driest provinces in which agriculture takes place. Recent experiments with these lower densities showed good results for groundnut and similarly also for soybean were obtained (Mabapa et al., 2010; Odhiambo et al., 2010). In addition, APSIM model run prior to this experiment to estimate potential density effect indicated only a small average yield increase of 132 kg/ha between a planting density of 7 plants/m² and 13 plants/m² for the period 1982–2012. Initial soil water was measured at the depths 0–15 and 15–30 cm. At harvest, dry matter and yield were determined by sampling of 10 plants. Samples were oven dried at 70 °C. Sporadic occurrence of aphids was controlled by spraying. At sowing 50 kg P/ha as Superphosphate was applied. No further fertiliser was used.

2.1.2. Cultivar × location trial

The second trial was conducted at two sites: at the University of Venda experimental site again and at one farmer's site in Makhado Municipality near Louis Trichardt (Fig. 1). Soil in Louis Trichardt is sandy loam over sandy clay. Organic carbon is 1.35% (Table 1). The experiment followed a randomised complete block design in factorial arrangement with three replicates: treatments were location and cultivar choice. The same two cultivars were planted at 6.67 plants/m² with a plot size of 3.5 × 3 m. Planting date was 1/11/2016 for University of Venda and 25/10/2016 for Makhado. Initial soil water was measured at the depths 0–15, 15–30 and 30–60 cm. Main phenological stages were recorded: emergence, date of first flowering (50% of all plants flowering), physiological maturity (90%). At flowering, eight

Table 1

Soil properties up to maximum rooting depth of University of Venda Research Station (Univen) and Makhado. CLL is crop lower limit. DUL is drained upper limit and KL is the potential fraction of available water that can be taken up from that layer.

Depth (Cm)	Bulk density (g/cc)	Air dry (mm/mm)	CLL (mm/mm)	DUL (mm/mm)	Saturation (mm/mm)	KL value (/day)	Organic carbon (%)	pH
Univen								
0–15	0.98	0.27	0.29	0.42	0.49	0.06	0.41	5.0
15–30	1.01	0.34	0.34	0.46	0.53	0.06	0.71	5.5
30–45	1.04	0.36	0.36	0.39	0.48	0.06	0.51	5.9
45–60	1.03	0.36	0.36	0.40	0.48	0.06	0.43	6.1
Makhado								
0–15	1.52	0.16	0.18	0.28	0.42	0.06	1.35	6.6
15–30	1.52	0.20	0.20	0.31	0.42	0.06	1.20	6.4
30–45	1.49	0.25	0.25	0.36	0.43	0.06	1.08	6.3

plants were harvested for each plot and dry weight of leaf and stem was measured. At harvest, total dry matter and yield were determined by sampling ten plants. Samples were oven dried at 70 °C. No fertilisers were applied due to the known history of high nutrient input in the past.

2.1.3. Statistical analysis of field trials

A two-way analysis of variance (ANOVA) was used each year to evaluate the effect of planting date and cultivar on biomass and grain yield. The same analysis was applied to the site x cultivar trial, where in addition leaf and stem weights at flowering were evaluated.

2.2. APSIM evaluation

APSIM (version 7.7) was used in this study (Holzworth et al., 2014). It is a widely tested and applied model in southern Africa: Recent applications are for instance, climate change effects on maize (Rurinda et al., 2015) or trade-offs in crop residue management (Baudron et al., 2015). A key feature is that APSIM has been especially used for evaluating grain legumes in southern Africa (Dimes et al., 2011; Mabapa et al., 2010; Odhiambo et al., 2010). It has also been applied to explore more complex interactions such as sorghum-cowpea intercropping (Chimonyo et al., 2016) or mucuna-maize and cowpea/groundnut-maize rotations (Masikati et al., 2014; Ncube et al., 2009; Robertson et al., 2005). As stated above, it is a process-based agro-ecosystem model, which simulates on a daily time step growth and development of a plant stand under given environmental conditions. It includes sub-modules for the major globally cultivated crops. The groundnut model is described by Robertson et al. (2002). Nitrogen (N) relations (demand, uptake and re-translocation) in APSIM are simulated similarly to the method used in the CERES model (Chen et al., 2016; Robertson et al., 2002). Three processes are driving the N uptake: ‘Mass flow’, ‘Active uptake’ (or Diffusion) and N fixation. It is assumed that mass flow is dominant. If N demand is not satisfied, N will be taken up via diffusion. Finally, if also this supply is not sufficient, N will be provided by fixation. Here, the model assumes optimal nodulation, and consequently no limitations of N fixation via pH or temperature (Chen et al., 2016). The N fixation capacity is only determined by plant size and water stress. The APSIM groundnut model can handle P limitations on plant growth, but has not been sufficiently tested yet. As we assume also no P limitations in our trial, we kept it switched off.

Beside plant growth and N relations, major soil processes as soil water dynamics are also simulated. Taking into account the importance of water in this highly variable environment, we shortly summarise the processes involved in the water balance (SOILWAT) in APSIM, for description of other process we refer to Holzworth et al. (2014) and for detailed testing for low rainfall cropping regions to Whitbread et al. (2017). The default APSIM version makes use of a tipping bucket approach, i.e. water flows from one defined soil layer to the next one. Water holding capacity of each layer is determined by crop lower limit (CLL, similar to permanent wilting point), drained upper limited (~field capacity) and saturation. Water flow through the profile is differentiated according to unsaturated and saturated moisture conditions. Water flow out of the profile is considered as drainage. Surface runoff is calculated using the USDA-Soil conservation curve. Potential evapotranspiration is simulated based on the Priestley-Taylor approach.

2.2.1. Model setup

APSIM was set up for all years (2013–2017) and the two sites where field trial data was available. Daily solar radiation, max and min temperature and precipitation for the Venda site was recorded close to the field. However, breakdown of the weather station caused gaps in the recordings for the season 2013/14 and 2016/17. Here we used data from available close station; Levubu for 2013/14 and Thohoyandou-Makwela for 2016/17 (6 km distance). For Makhado data from a nearby weather station was acquired (station name: Munuzwe Nuwco).

Table 2

Cultivar parameter adopted from the default cultivar Kangwana in the APSIM data base for ICGV 07396 and ICGV 99257.

Parameter	APSIM acronym	Unit	Kangwana (Default)	ICGV 07396	ICGV 99257
node phyllochron	y_node_app_rate	°C	56	56	66
Thermal time (TT) from flowering to start grain fill	y_tt_flowering	°C	300	440	440
TT from start grain filling to end grain filling	y_tt_start_grain_fill	°C	440	480	480

Soil texture and organic carbon (and at Univen also bulk density) was measured until rooting depth at both sites and were consequently used to determine plant available water capacity (Lower limit, drained upper limit and saturation, bulk density) using the pedo-transferfunctions from Saxton and Rawls (2006) (Table 1). For details of further parameterization the soil module we refer to Table A in the Supplementary material.

Initial water for model setup was measured for the layers 0–15 cm and 15–30 cm each year at planting (for 2016/17 also for 30–60 cm). For the layers below, fifty percent of field capacity was used. Initial N was not measured, but a conservative estimate of 33 kg N/ha for 0–60 cm was used (personal communication J. Odhiambo).

The two cultivars used in this study ICGV 07396 and ICGV 99257 are not part of the APSIM data base yet. We used the treatments early planting 2013/14 and the Univen 2016/17 experiment to calibrate these cultivars for phenology and leaf properties based on the existing cultivar Kangwana in the data base (Table 2). Phenology was matched to observed flowering and maturity. This was done by modifying the thermal time units required for the periods from flowering to start grain filling and from start grain filling to end grain filling. Interestingly, we found no differences in phenology between the two new cultivars; however, we observed in the field that ICGV 99257 was slower in canopy closure due to higher node phyllochron. This corresponded to lower amount of measured biomass at flowering. Hence, we modified node phyllochron for this cultivar from 56 to 66 °C. Node phyllochron describes the time between the appearance of one node and the next one in degree units. The APSIM groundnut model reacts to water deficits, among others, by delayed phenology. This specific feature was built in because of observations made for the Virginia groundnut types (APSIM, 2017). We could not observe this phenomenon in the Spanish type cultivars in our trial, and consequently, switched it off.

2.2.2. Model evaluation

Model evaluation was done against the treatments and years' data available from the experiments. Exceptions were the treatment early planting in 2013/14 and the 2016/17 experiment at Univen, which were as stated above used for calibration. Thus, these were excluded from the model evaluation. Model results were statistically compared with observed findings using standard methods: root mean square error (RMSE), Pearson's correlation coefficient and index of agreement (IA, Willmott et al., 2012).

2.3. Simulation experiment

2.3.1. Model setup

To test the hypotheses of this study at a larger spatial scale over a longer period, a simulation experiment using APSIM for all arable rainfed land in Limpopo at a 0.01° grid for the period 1955–2010 was conducted with the same design as the planting date by cultivar field trials in Venda; three levels of planting date and two cultivars resulting in six treatments: Two groundnut cultivars were chosen (the calibrated

ICGV 07396 and ICGV 99257). In addition, as we found no major difference in phenology between these two cultivars, we selected a local early-maturing groundnut cultivar from the APSIM data base called Kangwana, which has been calibrated for Limpopo (Odhiambo et al., 2010). This was done to enable us to explicitly test our hypothesis that an early cultivar can be beneficial (Table 2).

Planting date was early (20-October–19-November), mid (20-November–20-December), and late (21-December to 20-January). Planting in each time period was triggered by amount of rainfall > 10 mm and at least three rainfall days, and minimum allowable available soil water for the profile was at least 20 mm reflecting local rules of thumb. If criteria were not fulfilled during that period, the crop was sown at the last day of that planting window. Planting density was 6.67 seeds/m². The simulation runs were conducted for the years 1955–2010. Surface organic matter set to 500 kg/ha of maize reflecting the standard rotation of the region. Following Sennhenn et al. (2017), surface organic matter and soil N (based on the 33 kg N/ha for the 0–60 cm in the evaluation run 10 kg N/ha were added for the deeper layers, which resulted in 43 kg N/ha for the whole profile up to 180 cm eliminating the effect of N on plant growth) were reset every year at the beginning of the growing period (15-October). Soil water was not reset to include the effect of climate conditions during the fallow period (as in similar studies; for instance Hoffmann et al., 2017).

2.3.2. Gridded climate, land use and soil data

Gridded weather data was sourced from the daily global meteorological forcing dataset for land surface modeling (Sheffield et al., 2006). The dataset was developed by adjusting reanalysis data to observations of climate research unit's monthly temperature and precipitation totals. The reanalysis was obtained from the National Centers for Environmental Prediction/National Center for Atmospheric Research (Kalnay et al., 1996). The variables included minimum temperature, maximum temperature, global solar radiation and precipitation at 0.25° resolution. This data has been used as well for hydrological large scale simulation analysis for South Africa (Estes et al., 2014). Land use data were extracted from the South Africa National Land-Cover Change Map 2000 (Schoeman et al., 2013).

Soil data was used from Leenaars et al. (2015), which provides hydrological soil properties including rooting depth and chemical properties (organic carbon and pH), as input to APSIM. Parameterization of the soils for APSIM was done according to Dalglish et al. (2016). Soil data was available at a resolution of 1 km in Lambert's azimuthal equal area cartographic projection centered at 5°N and 20°W. Data were converted, i.e. re-projected and resampled to WGS84 geographical coordinates at 0.01° resolution using nearest neighbor resampling to match the land use dataset. Simulation runs were conducted at this resolution for all arable cropping land according to the map of Schoeman et al. (2013). Overall, there were 14,739 grid cells to cover Limpopo province. Since we considered nine management options, this resulted in 132,651 simulation runs times 55 years. To reduce computing time, we run these simulations in parallel on the Goettingen Research Campus cluster, which reduced the processing time from ~46 days (using one core of one computer) to less than half a day. Finally, we compared for each grid cell planting date and cultivar choice divided into El Niño, neutral years, and La Niña years, which were defined as mentioned above by the ONI (NOAA Climate Prediction Center, 2017).

3. Results

3.1. Field trial results

Field trial results showed that yields varied significantly between years and between treatments (Figs. 3 and 4). Yield ranged from a maximum of 2500 kg/ha achieved by cultivar ICGV 07396/early planting in season 2013/14 to zero yields in 2014/15 and 2015/16,

when plants did not germinate under the late planting, no matter the cultivar choice (Fig. 3). Hence, planting date had a strong effect on productivity; clearly, early planting was superior to delayed planting across all years. In terms of cultivar choice, ICGV 07396 performed generally better than ICGV 99257 (Figs. 3 and 4), except for the cases of extended dry spells as occurred for mid and late planting in 2014/15 and 2015/16. For mid and late planting in the season 2013/14 (Fig. 4) productivity was low. ICGV 07396 showed a faster growth rate indicated by higher stem and leaf dry matter at the flowering stage, although not significant at $\alpha = 0.05$ (Fig. 3). Surprisingly, we did not observe differences in phenology and consequently growth duration was not different between the cultivars (see Supplement Fig. E). Groundnut cultivars had similar yield levels at Makhado and Univen. At both sites high total biomass production was achieved (> 6000 kg/ha), while grain yields remained below 2000 kg/ha.

Grain yield in general followed total biomass at harvesting; just under dry conditions as in the seasons 2014/15 and 2015/16 for mid planting harvest index was distinctly reduced (Fig. 3).

A key feature of the field trial was the strong variability of productivity between years, which coincided with the variability in rainfall amount and its seasonal distribution. In 2016/17 there was more than 1300 mm rainfall during the growing season, and also for the early planting in 2013/14, groundnut received close to 1000 mm. Mid and late planting led to groundnut cultivation periods with less rain (lowest value was 200 mm in 2014/15 for mid-planting, and even less for late planting, which was not possible anymore (Supplement, Fig. S5).

3.2. Crop model evaluation

Overall, biomass was predicted satisfactorily by APSIM indicated by the IOA of 0.73 and a RMSE of 823 (kg/ha) against a mean of 3255 (kg/ha) (Fig. 5). For grain yield, we found a relatively high RMSE of 331 kg/ha taking the mean of only 663 kg/ha into account. However, the IOA of 0.62 was still in a good range. For both, biomass and grain yield, the Pearson correlation coefficient (0.62 and 0.73) was reasonable indicating that the seasonal and treatment effect were captured.

3.3. Simulation experiment

Generally, yield averaged across all management practices differed significantly between climatic conditions. Highest yields were achieved in La Niña (median 996 kg/ha across all grid cells) years and lowest in El Niño (654 kg/ha across all grid cells) (Fig. 6). However, yields varied strongly within Limpopo: higher yields of more than 1000 kg/ha in the South-East and East and below 1000 kg/ha in the West under neutral climatic conditions. In El Niño years, yields in the West dropped below 500 kg/ha, while in the East, yields of more than 1000 kg/ha were still achieved. Maximum yields of above 2000 kg/ha were attained in the La Niña years for few grid cells in the center of Limpopo. The spatial differentiation of simulated yields largely followed the spatial distribution in rainfall (Figs. 1 and 6).

In terms of yield variability of each grid cell we found strong variability for the low rainfall regions of the North-West and lower variability in the East and South indicated as coefficient of variation (Fig. 7). Thus, seasonal yield variability and associated risk of poor yields was higher in the North-West.

Taking all grid cells in Limpopo, it appeared that early planting is one important key to achieving high production. Under all three (ENSO-related) climatic modes yield declined from early to late planting (Fig. 8). This pattern was more dominant for the later maturing cultivars ICGV 99257 and 07396 in comparison to earlier maturing cultivar Kangwana. It is also important to note that highest yields (i.e. above 3500 kg/ha) were only achieved by the later maturing cultivars under neutral and La Niña conditions. In El Niño years, grain yield remained below 3000 kg/ha even at the most productive sites.

While the combination of early planting and cultivar ICGV 03796

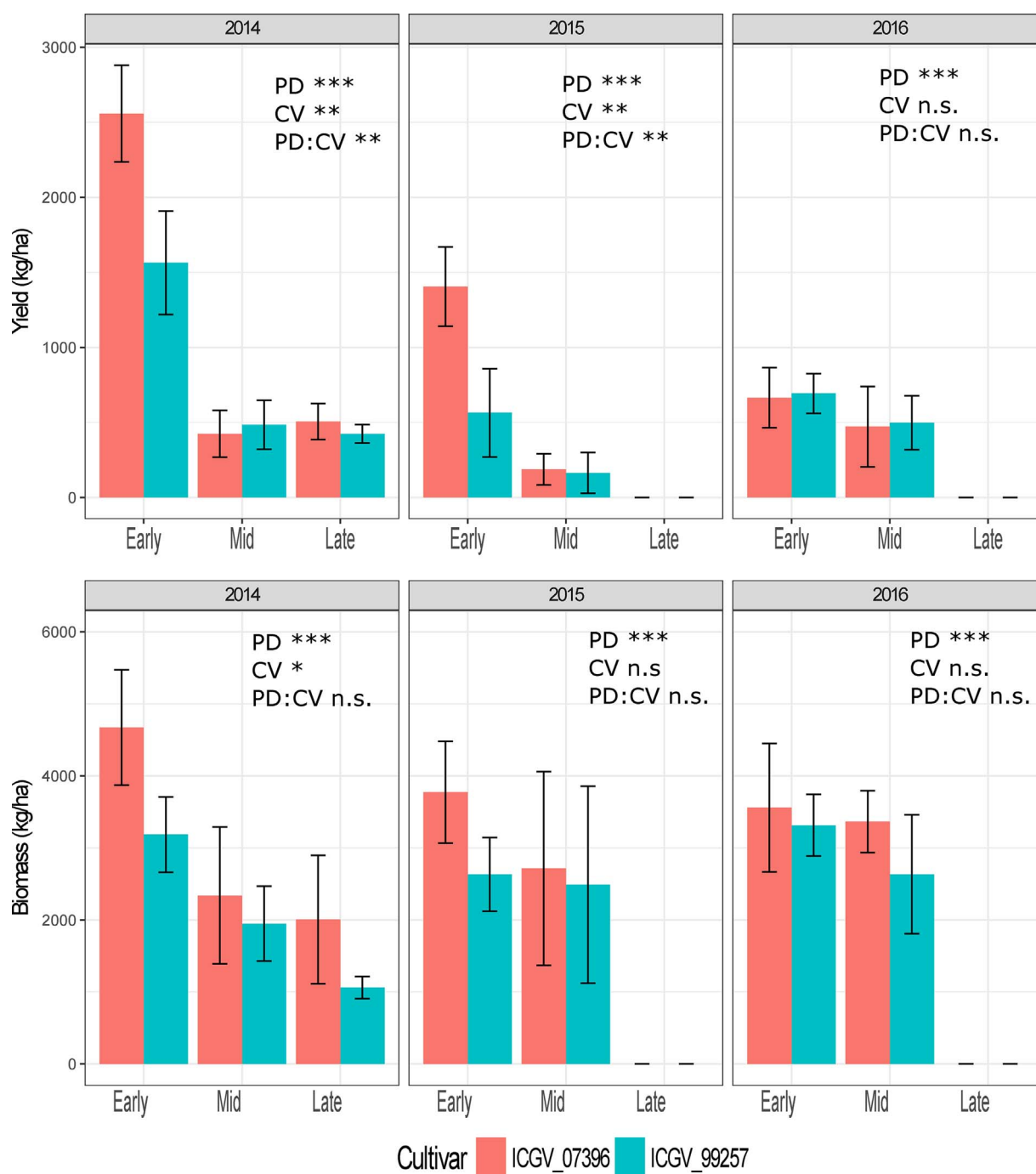


Fig. 3. Mean (bar) and standard deviation (error bars) of grain yield (a) and total biomass (b) as affected by planting date (early, mid, late) for two groundnut cultivars (ICGV 07396 and ICGV 99527) in Univen for the seasons 2013/14, 2014/15 and 2015/16 indicated by the year of harvest above each plot. Significance effect of management – practice planting date (PD), cultivar (CV) and its interaction (PD:CV) – is indicated for each year separately at $P \leq 0.001$ ‘***’, $P \leq 0.01$ ‘**’, $P \leq 0.05$ ‘*’ and non-significant (n.s.) at > 0.05 .

was superior looking at the whole province (Fig. 8) and across the different climate modes, looking closer at the nine management options (cultivar choice x planting date) revealed some site and season specific differences (Fig. 9). Overall, early planting x ICGV 07396 can be recommended for 73% of all grid cells for El Niño years, for 72% for neutral conditions, and for 69% for La Niña conditions. For El Niño years, the combination of early planting and the early maturing cultivar Kangwana achieved the highest yield in the dry regions (12% of grid cells), and Kangwana x mid planting for 8% of all grid cells. Under neutral and La Niña conditions, also mid and x late planting in combination with cultivar ICGV 07396 led to best performance (12% and 10% for neutral, and 9% and 19% for La Niña). Finally, comparing the

yield differences between site- and season- specific management recommendations and the standard recommendation (i.e. early planting x cultivar ICGV 07396), we found yield increases of more than 20% for some sites in El Niño years, for neutral years almost no differences in yield and in La Niña years we found 6–10% increases for wetter sites located in the center of Limpopo (Fig. 10).

4. Discussion

Overall, we found that early planting and cultivar ICGV 03796 was the most productive management combination across sites and across climate condition in the simulation experiment, i.e. 73% of all grid cells

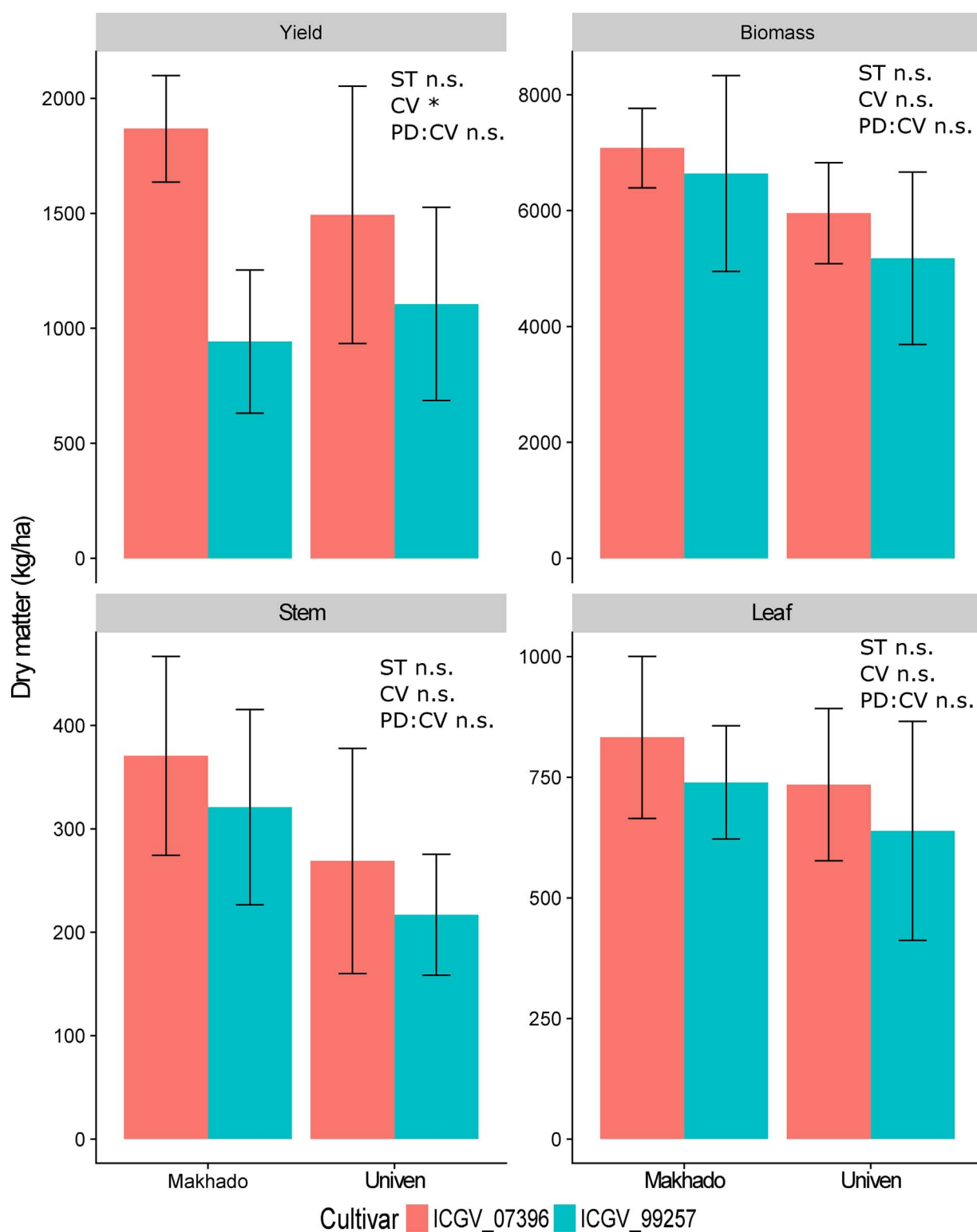


Fig. 4. Grain yield and total biomass at harvesting, leaf and stem at flowering as affected by site (Makhado, Univen) for two groundnut cultivars (ICGV 07396 and 99527) for the season 2016/17. Significance effect of study site (ST), cultivar (CV) and its interaction (ST:CV) is indicated at $P \leq 0.001$ as '***', $P \leq 0.01$ as '**', $P \leq 0.05$ '*' and non-significance (n.s.) at $P > 0.05$.

for El Niño years, for 72% for neutral conditions, and for 69% for La Niña (Fig. 9). We found the same pattern in the field trial (Fig. 3). However, a closer look at the spatial pattern of the simulation analysis revealed that, for the general drier regions of Limpopo as hypothesized, an early maturing cultivar can lead to more than 20% yield increase (Fig. 10). In addition, in El Niño years early planting is crucial for most

sites (85% of all grid cells) in comparison to neutral (74%) and La Niña (70%) years. Hence, due to the heterogeneous climate x soil conditions in Limpopo a detailed spatially and temporally analysis as shown in this study was necessary to identify suitable adaptation options. The approach and the implications are discussed in more detail below.

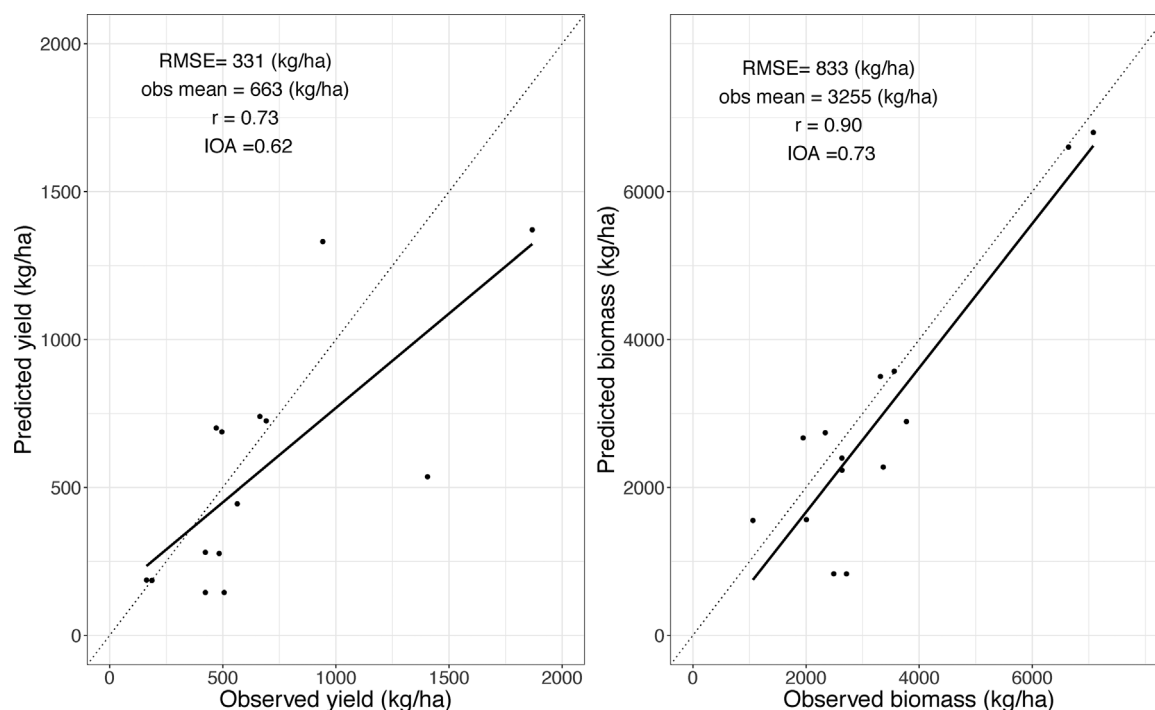


Fig. 5. Predicted and observed grain yield and biomass. Statistical model evaluation has been done using root mean square error (RMSE), Pearson correlation coefficient (r), and Index of Agreement (IOA) according to Willmott et al. (2012). Additionally, the observed mean (obs mean) is given. Dashed line is 1:1 line and straight line is regression line. Data for these two figures is presented in Table B in the Supplementary material.

4.1. Model evaluation

Before discussing the management effects on groundnut productivity, we need to put the gridded crop modelling approach applied here in the context of model evaluation in general, and highlight some important aspects. When comparing observed and predicted grain yields at field scale, it was shown that the model overall was capable of capturing seasonal dynamics and the effect of management practices as indicated by IOA of 0.62 and r of 0.73. However, for the absolute values there was a relatively high RMSE of 331 kg/ha. Such high RMSE can occur when the observed mean yields are quite low, as found in many low rainfall regions (see for instance Hochman et al., 2009; Hoffmann et al., 2017), including southern Africa (Ncube et al., 2009; Rurinda et al., 2015), and was still found to be acceptable in those studies. Biomass was simulated with a normalized RMSE of just 26%. Also the Pearson's r (0.9) and IOA (0.73) indicated good agreement between model predictions and field trials observations.

Overall, this model evaluation exercise showed that APSIM is able to reproduce the overall effects of management on groundnut productivity and different sites and seasons in a reasonable range. This, in conjunction with other literature confirming the capability of this model to successfully simulate cropping systems in southern Africa (Baudron et al., 2015; Ncube et al., 2009; Robertson et al., 2005; Rurinda et al., 2015; Whitbread et al., 2010) might serve as basis to use the model in simulation experiments.

4.2. Management adaptation

Generally, as expected, there was a clear season-dependent pattern for yields, with a decline from La Niña via neutral to El Niño years (Fig. 8) (as observed for other crops elsewhere (Iizumi et al., 2014)). Spatially, yields reflected the rainfall gradient from humid subtropical with high average growing season rainfall of more than 600 mm to the semi-arid regions in the north-east of the province with rainfall below 400 mm (Figs. 1 and 6). The coefficient of yield variation as indicator for seasonal climate-induced risk was highest in these dry regions and

lowest in the wet regions. These findings confirm common knowledge that productivity in Limpopo is closely linked to rainfall pattern (Akpulu et al., 2011).

Across climate modes, we found the overall management combination early planting x cultivar ICGV 07396 as the most successful in terms of productivity. In El Niño years this option was the most successful for 73% of all grid cells, in neutral years in 72%, and for La Niña for 69% (Fig. 9). This finding of the spatial simulation analysis corresponded to our field trial results (Fig. 3). It showed that for a large range of sites, a standard practice appears to be sufficient for high production and season-specific adaptation to climate conditions was not really helpful for those management practices we considered in the analyses account. However, a spatially differentiated analysis demonstrated that, for a substantial amount of sites, there were still promising adaptation options: In El Niño years the early maturing cultivar Kangwana actually yielded higher than the ICGV 07396 for 20% of all grid cells (12% when planted early and 8% when mid planted) (Fig. 9). Changing in these sites in El Niño years from a standard recommendation to Kangwana early respectively mid planting resulted in up to more than 20% higher yields (Fig. 10).

The explanation might be that in El Niño years, water was by far the most controlling factor. Under these water scarce conditions, it was naturally most important to capture as much rainfall as possible. And indeed early planting, resulted in the higher amounts of rain captured (305 mm for early planting, 281 mm for mid planting and 233 mm only for late planting averaged across the region). In drier regions, the combination as we hypothesized of early planting and an early maturing cultivar escaping the dry spell consequently led to the highest production. Our simulation and field trial results revealing that early planting is crucial for high yields underlined the importance of this management practice (Rurinda et al., 2015; Waha et al., 2013). Our findings confirmed those from studies on other crops for southern Africa indicating the importance of early planting in November (for maize for instance: Phillips et al., 1998) and also the agricultural extension advice for groundnut for Limpopo (Department of Agriculture, 2010). This strategy however, is practically constrained by the need to

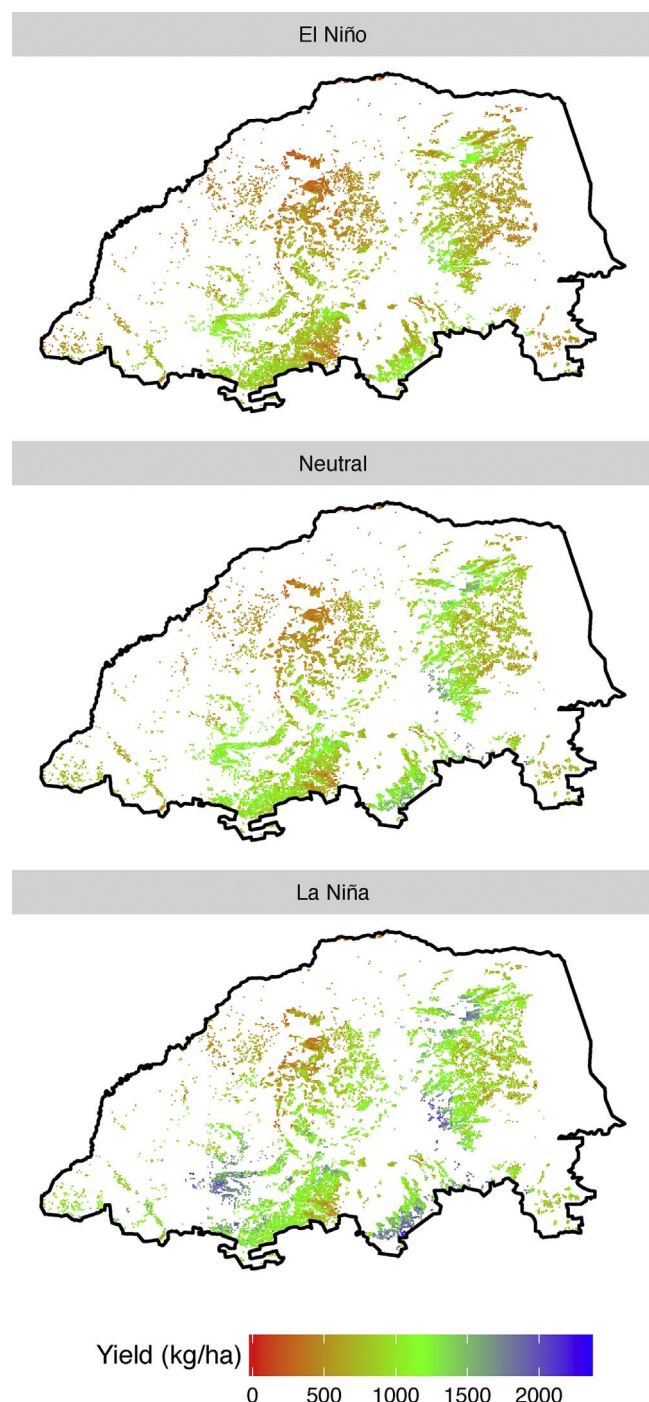


Fig. 6. Simulated median yield across management practice for the climatic conditions: El Niño, neutral and La Niña years. Simulation period was 1955–2010.

have sufficient soil water available for planting. That means, initial water content stored in the soil should be sufficient for the crop to also overcome a possible short dry period immediately after planting (Raes et al., 2004). Thus, early planting comes with a risk in case not sufficient water was stored in the soil. Supplementary irrigation at planting, where possible, and/or short range (seven to fourteen days) weather forecasts, might be of potential use to minimize the risk in this situation.

Under high rainfall conditions (neutral and La Niña years), we also found out that later planting dates by cultivar ICGV 07396 lead to reasonable yield. This was obviously caused by a higher water supply across the season (and hence less water limitations) and a longer

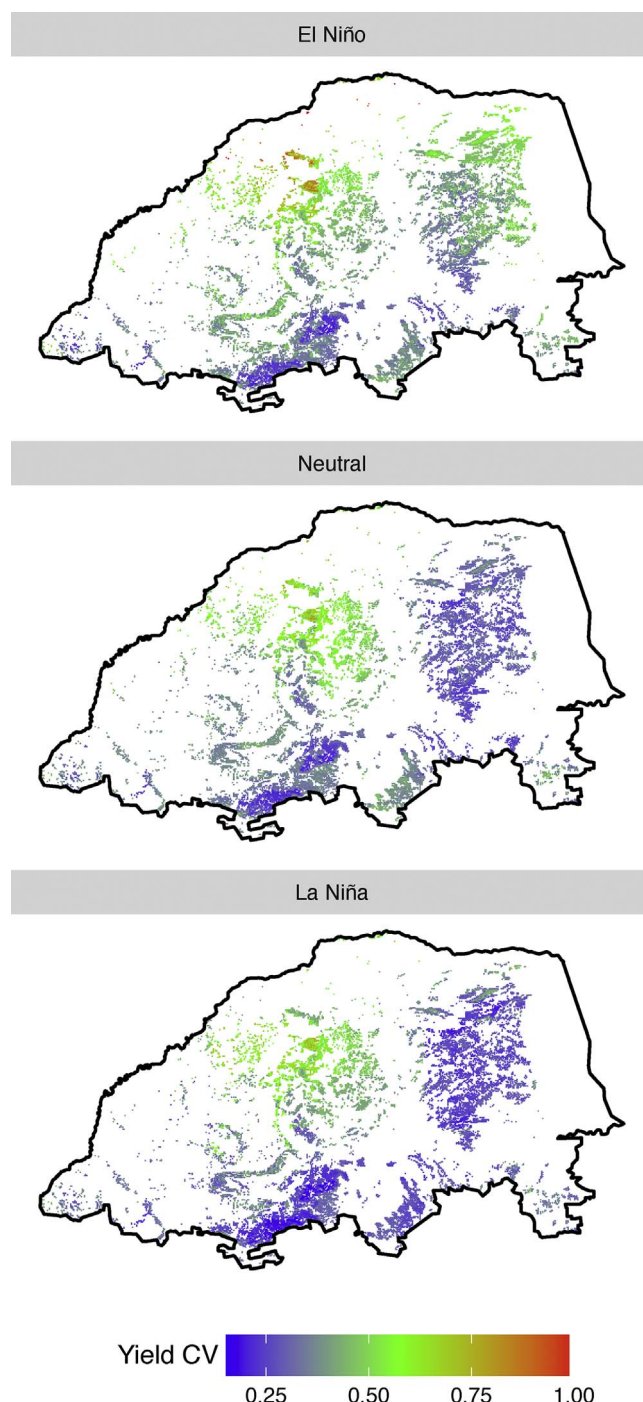


Fig. 7. Simulated coefficient of variation (CV) of yield across management practice for the climatic conditions: El Niño, neutral and La Niña. Simulation period was 1955–2010.

growing period (cultivar choice: for ICGV a duration of 128 days and for Kangwana a duration of 122; planting date: a duration of 127 days for early planting, 130 for mid and 136 days for late planting). This result was in line with the findings by Vadez et al. (2017) for Zimbabwe, where longer growing duration of groundnut was positively related to productivity.

Results from this study demonstrate that indeed season-specific management could offer opportunities to maintain productivity under extreme climatic conditions as during El Niño years. This option, however, requires a site-specific dimension as well and is restricted to 27% of all grids in the province. In total, such a site specific adaptation would improve productivity in El Niño years for Limpopo by 2.9%, in

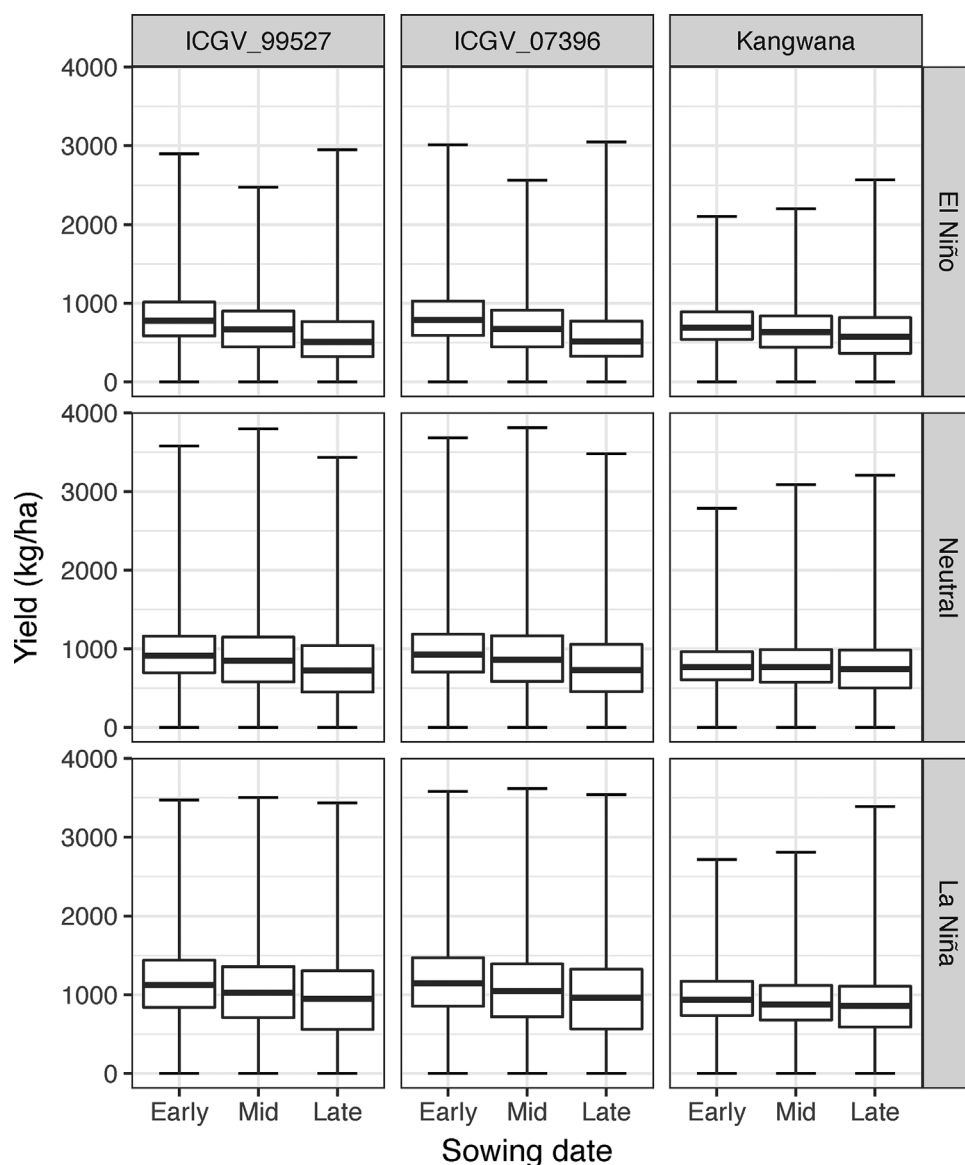


Fig. 8. Boxplots indicate the effect of planting date and cultivar choice on simulated yield for each climate condition (El Niño, neutral and La Niña) across all grid cells. Whiskers indicate the 0 and 100% quantiles. Simulation period was 1955–2010.

neutral years by 1.5% and in La Niña by 2.7% (Fig. 10).

While these numbers could give the overall impression of low capacity to buffer the effect of climate extremes by management, for some vulnerable sites, benefits are clearly higher: an early maturing cultivar and modifying planting date could be indeed a specific adaptation, which can result in > 20% higher yields (Fig. 10). Identifying such sites requires, however, a spatially explicit approach as presented in this study.

4.3. Improvement requirement

We combined field trials and gridded simulation modelling to evaluate planting date and cultivar choice as adaptation strategy to climate variability. While we found clear patterns as discussed above, there are limitations, which need to be considered: As mineral fertilization is not common in smallholder sites in Limpopo (Odhambo and Magandini, 2008) phosphorous limitations might be possible for those sites, which are neglected in our analysis. Biotic stresses were not included in the simulation study. However, it is known that in wet years (La Niña) early leaf spot (*Mycosphaerella arachidis*) is a challenging disease for groundnut in South Africa (Department of Agriculture, 2010). This might counteract for some sites the positive effects of La

Niña years in comparison to neutral and drier years when no prevention is conducted. Similarly, insects such as aphids can become a major limiting factor. In our trial we eliminated the occurrence of them right from the beginning. Termites can become a pest, in particular in dry years. Hence, it is important to underline that these simulated yields in our study are attainable only when biotic stresses are controlled. The lacking capability of most crop simulation models (here APSIM) to include these effects of biotic stress on yield is widely known, but little has been done so to overcome this – though some new initiatives have been emerging in the framework of ag modelling networks AgMIP and MACSUR. Another limitation was that we only evaluated three groundnut genotypes. Testing of a much wider range of genotypes would be required to explore in more detail the scope and utilize the potential of genotypic adaptation as shown for other crops (see, e.g. Tao et al., 2017). Traits related to phenology and finally growth duration might be key in this regard as suggested in the simulation experiment by comparing the two cultivars ICGV 07396 and Kangwana. Surprisingly, we found in our field trial contrary to our expectation no difference. One explanation could be that in our trials we found higher amount of rainfall as crops typically experience at the ICRISAT campus at Hyderabad in India, where these cultivars are originating from. Water deficiency could have affected the development of these cultivars

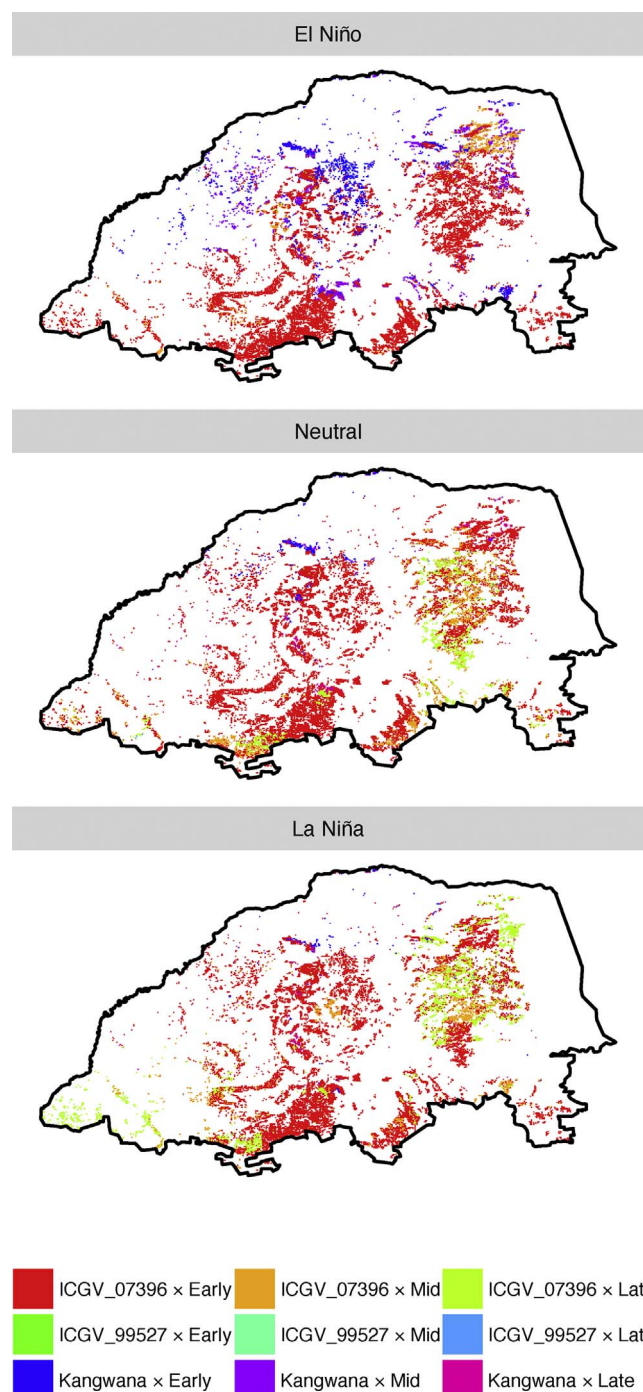


Fig. 9. Map shows the combination of management practices (cultivar choice x planting date), which resulted in the highest median yield for each grid cell for the specific climate mode (i.e. El Niño, neutral, La Niña years). Simulation period was 1955–2010.

differently (Ong, 1985). Another important factor, which requires more attention, is planting density: Vadez et al. (2017) found using a simulation model for Southeast (Kenya to Zimbabwe) and West Africa modifying planting density a promising adaptation option. For Limpopo planting densities of 6.67 plant/m² are rather conservative, and it might be of interest verifying whether higher densities could potentially increase productivity in wetter years.

With respect to the grid based simulation approach, uncertainties linked to model input for such spatial analysis is well known (Müller et al., 2017). For instance, soil data choice plays an important role for simulated yields (Folberth et al., 2016). Here, we used a specific

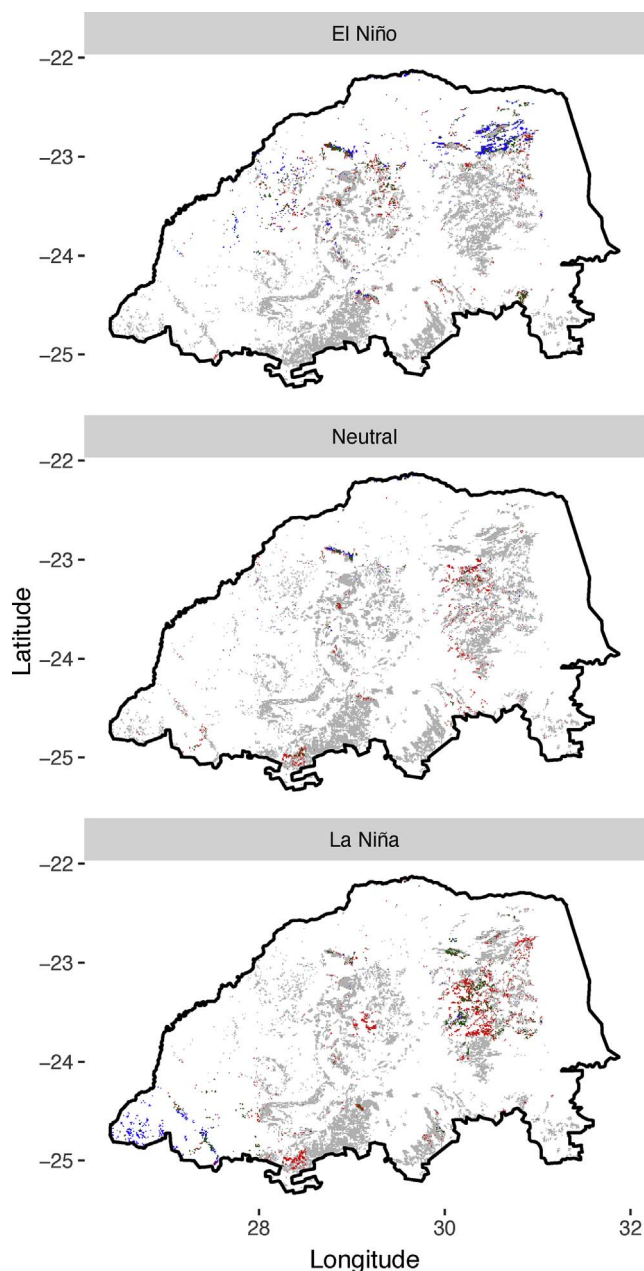


Fig. 10. Map shows increase in percentage for each grid cell if best performing practice for a given site and climate mode (i.e. El Niño, neutral, La Niña years) is used instead of standard management (ICGV 07396 x early planting). Simulation period was 1955–2010.

adopted data set for Sub-Sahara Africa based on the original International Soil Reference and Information Centre (ISRIC) data, which provided explicitly plant rooting depth (Leenaars et al., 2015). However, comparing simulation analysis with different soil data sets and also different pedotransfer functions could help to better quantify the uncertainties related to soil input data.

5. Conclusion

This study made use of field trials and grid based simulation analysis to evaluate possible adaptation of groundnut cropping to climate extreme events such as El Niño in Limpopo province (South Africa). We found early planting to be a successful strategy across different weather

conditions. The cultivar ICGV 07396 performed overall best. In El Niño years, however, an earlier maturing cultivar, Kangwana, was beneficial for drier regions with yield increases of more than 20% for some sites. Hence, this study demonstrated that management recommendations at provincial levels need explicitly a more detailed site and season specific dimension. Further studies should evaluate a much wider range of genetic material to identify the most appropriate adaptation option to climate extremes.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.fcr.2018.01.019>.

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